A voice agent that builds a voice agent.

Dark software factories with Rasa Mantle, Deepgram and the OpenAI Agents API

“Build me an assistant for my next flight.”

A checked build.
A working preview.
A conversation you can try.

“Will I make my meeting?”

A question that needs more than a flight number.

FLIGHT · ILLUSTRATIVE

Landing 17:00

Computed estimate
Check the source timestamp



WEATHER · ILLUSTRATIVE

Light rain

Arrival-time forecast
A forecast is not an observation



MEETING · COMPUTED EXAMPLE

30 minutes late

17:00 landing + 90 min transfer
18:00 meeting, same local date

CURRENCY

Daily reference

Publication date and source
Unavailable when the source fails

EUR / USD

The screen makes the answer inspectable.

A useful teaser shows both a success and a failure.

01

Ask by voice

Choose a flight. Give a meeting time. Hear a brief answer.

02

Change one assumption

Increase the transfer allowance. Watch only the timeline change.

03

Lose one source

Show weather as unavailable. Keep the other cards useful.

Most of today belongs to your group.

110 minutes building, 15 minutes meeting other builders.

Demo	Concepts	Groups	Lab A	Meet	Lab B	Share
10	20	10	55	15	55	15

Minutes from start to finish: 180. The guide lets you work at your own pace.

Two agents have different jobs.

The factory creates the product. The traveller uses the product.

BUILD TIME · TARGET ARCHITECTURE

You describe a product

Factory voice assistant → build worker → independent checks → released preview

RUN TIME · TRAVEL SCAFFOLD

You ask about a trip

Travel voice assistant → source tools → validated data → visual cards

The attached scaffold and builder currently run separately. Their automatic release bridge is extension work.

Each part has a clear responsibility.

The cards and factory controller are application code.

Part	Its job
Rasa Mantle	Run the conversation, skills and tool constraints
Deepgram	Turn speech into text and text into speech
OpenAI Agents API	Run a durable builder session and its managed harness
Factory controller	Scope work, verify a candidate and decide release
Source adapters + visual UI	Fetch data and render trustworthy cards

The Agents API runs the builder session.

Python’s OpenAI client is the transport used by this implementation.

```
from openai import OpenAI

client = OpenAI()
client.beta.agents.sessions.create(
    agent={...},
    environment={...},
    input=task,
    stream=True,
)
```

Four things to follow

Agent configuration
Execution environment
Session identity
Events and final status

Conceptual excerpt, not a complete runnable script.
The separate `openai-agents` framework is the Agents SDK.

MCP is the connection to your application.

A tool call needs a contract, a result and a failure path.

01

Mantle skill

Requests a bounded operation

02

MCP service

Validates arguments; starts a job

03

Build worker

Uses the Agents API

04

Job result

Returns status and evidence

A brief becomes a testable request.

“Make it better” leaves the builder guessing.

PRODUCT REQUEST

“Show only two currencies, with the reference date. If rates fail, say they are unavailable.”

ACCEPTANCE EXAMPLES

- Four input rates produce two rows.
- The publication date is visible.
- Daily rates never say “live”.
- Source failure produces no invented rate.

The builder cannot grade its own homework.

An independent verifier checks the proposed artifact.

BUILDER WORKSPACE

Task + product files

Allowed changes only
Development tests available

candidate

VERIFIER WORKSPACE

Protected acceptance

Independent result
Task and artifact fingerprints

A copied directory is not an operating-system sandbox. Protect execution as well as files.

A completed session still needs a decision.

A setup failure is different from a bad product.

01

ACCEPT

The candidate passed the specified checks. Release checks still follow.

02

REJECT

A product requirement failed. Change the proposal and try again.

03

NOT_READY

A dependency, account or verifier failed. No product verdict exists.

Dark factories need an explicit release policy.

People define the standards and own exceptions.

01

Requirements

People specify acceptable behavior



02

Build + checks

Automation produces evidence



03

Release gate

Policy permits or stops a release



04

Monitor

Failures trigger rollback or escalation

Your group owns one small change.

Driver, requirements owner, evaluator. Rotate after the break.

01

Write examples

Choose a visible feature. Predict the failure cases.

02

Run the factory

Keep the task, candidate and recorded verdict together.

03

Show the result

Demonstrate a rejection and explain what you released.

One guide, two routes.

Choose the route your machine can run today.

Route	Requirements	Outcome
A · Travel + code	Mac for code gates; licence; provider keys for live voice	Voice scaffold, builder, rejection, local card integration
B · Factory method	Python 3.12; no provider accounts	Acceptance, recovery and duplicate suppression
Work together	One prepared machine per group	Everyone writes examples and evaluates results

Lab A: a brief and a first candidate.

55 minutes. Open PARTICIPANT-GUIDE.md.

01

10 min

Brief + failure examples

→

02

15 min

Travel experience or fallback

→

03

25 min

One bounded builder attempt

→

04

5 min

Record evidence

A source label changes what the answer means.

The same number can have very different authority.

Card / value	Meaning	What to show
Aircraft position	An observation at a timestamp	Source and observed time
Arrival time	Computed estimate	Method and uncertainty
Weather	Forecast for a place and time	Forecast time and provider
Transfer	Planning assumption	Editable allowance
Currency	Daily reference rate	Publication date; never live trading

Meet another group.

15 minutes. Bring back one failure your tests missed.

“What would make your assistant confidently wrong?”

Trade an example. Explain how you would detect it. Swap roles when you return.

Lab B: rejection, integration and handoff.

55 minutes. Finish with evidence you can show.

01

20 min

Reject a misleading candidate

02

25 min

Integrate and inspect output

03

10 min

Prepare a two-minute demo

Release the exact artifact that passed.

A changed base or changed candidate invalidates earlier evidence.

TESTED CANDIDATE

Digest A

Task + verifier + results



RELEASE CHECK

Still digest A?

Same base? Smoke tests pass?



RUNNING PRODUCT

Verified version

Preview + rollback reference

The full voice factory also needs a start-and-smoke-test release worker.

Voice has its own acceptance tests.

Passing text tests leaves important behavior untested.

01

A short reply

Does “yes” become the right confirmed action?

02

An interruption

Does playback stop while the user changes the request?

03

A failed source

Does speech explain the problem while the board keeps useful data?

Your demo starts with the requirement.

Two minutes per group. Show the failure as well as the success.

30 s

The request

60 s

The rejected case and the result

30 s

The next test

Include your mode: LIVE, SCRIPTED or REPLAY. Name any step you did not run.

The next step is the complete build-to-voice journey.

One spoken brief should lead to a checked, usable preview.

ALREADY REPRESENTED IN THE PACK

Travel scaffold + separate builder

Source tools, cards, scoped changes, acceptance and local integration.

REMAINING INTEGRATION

Voice factory + durable jobs + release

Build tool, status, cancellation, exact-artifact launch and voice smoke tests.

One question to take back to work.

Your answer determines where to automate next.

“What evidence would let us release this change without reading every line?”

Start with one bounded change. Keep the rejected cases.